CURRICULUM VITAE

General Information

Name: Patpicha Arunsan, PhD

Rank: Lecturer

Date of birth: August 6, 1987

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Academic Credentials

- Lecturer at Faculty of Medicine, Vongchavalitkul University, Nakhon Ratchasima, Thailand (2025-present)
- Full-time Researcher at Institute of Research and Development, Suranaree University of Technology (2021-2025)
- Researcher at Parasitology Department, Faculty of Medicine,Khon Kaen University (2020-2021)
- Doctor's degree in Parasitology Department, Faculty of Medicine, Khon Kaen University (2020)
- Postdoctoral at Faculty of Medicine, Chulalongkorn University (2019-2020)
- Study nurse at The Infectious Research Unit, Faculty of Medicine, Khon Kaen University (2013-2015)
- Master's degree in Parasitology Department, Faculty of Medicine, Khon Kaen University (2014)
- Bachelor's degree in Nursing (Second class honors), Khon Kaen University (2010)

Course Training & Scholarships

- Doctoral student of RGJ to research at The George Washington University, Washington, D.C., USA (2018-2019)

- Higher Education Research Promotion and National Research University Project of Thailand, Office of the Higher Education Commission, through the Health Cluster (SHeP-GMS), Khon Kaen University.
- Thailand Research Fund through the Royal Golden Jubilee (RGJ) PhD Program (grant no. PHD/0111/2557).

Research Interests

- Herbal activity
- Anthelmintic drugs
- Nanotechnology
- Products
- Parasitology
- Parasitic diseases
- Liver Fluke
- Cholangiocarcinoma
- Molecular biology
- Gene editing
- CRISPR/Cas9
- Cell and tissue engineering
- Cell biology
- Tissue biology

International Conferences

1. **Patpicha Arunsan**. Eco-Friendly Synthesis of Capsaicin Nanoparticles: A Challenge to Enhance Pain Relief and Safety Profiles. The international conference; CRSingapore 2025, 19-21 February 2025, National University of Singapore (NUS), Singapore.
2. **Patpicha Arunsan**. Prevalence of oral parasitic protozoan *Entamoeba gingivalis* and risk factors among the periodontal disease patients in northeastern Thailand. The International conference on Parasitology (ICP) 2023, 9-10 November 2023, Khon Kaen, Thailand. And got award **"the outstanding presentation"**.

3. **Patpicha Arunsan**. Diminished hepatobiliary disease during infection with CRISPR/Cas9-gene-edited liver flukes, *Opisthorchis viverrini*. Universal Academic Cluster InternationalAutumn Conference, 16-18 October 2019, Tokyo, Japan.
4. **Patpicha Arunsan**. Diminished hepatobiliary disease during infection with CRISPR/Cas9-gene-edited *Opisthorchis viverrini* liver fluke. GW Cancer Center Basic Sciences Retreat, 27April 2018, The George Washington University, USA.
5. **Patpicha Arunsan**, Apiporn Suwannatrai, Thidarut Boonmars, Jariya Umka, Songpol Oopachitakul, and Smarn Tesana. Localization of the cystogenous glands of *O. viverrini* cercariae. The 13th workshop of the Regional Network for Asian Schistosomiasis and OtherHelminth Zoonoses (RNAS+) on 24-26 October 2013, Orchid ball room 2, Pullman RajaOrchid Hotel.
6. **Patpicha Arunsan**, Apiporn Suwannatrai, Thidarut Boonmars, Jariya Umka, Songpol Oopachitakul, and Smarn Tesana. Localization of the cystogenous glands of *Opisthorchis viverrini* cercariae. The 2nd symposium of Specific Health Problem in Greater Mekong Sub-region (SHeP-GMS) on 29 March 2013, Auditorium 1, Faculty of Medicine, Khon KaenUniversity.
7. **Patpicha Arunsan**, Chalermlap Donthaisong, Apiporn Suwannatrai, Thidarut Boonmars, JariyaUmka, Songpol Oopachitakul, and Smarn Tesana. Localization of the cystogenous glands of *Opisthorchis viverrini* cercariae. Joint International Tropical Medicine Meeting 2012 & The7th Seminar on Food and Water-Borne Parasitic Zoonoses on 12-14 December 2012, Centara Grand & Bangkok Convention Centre at CentralWorld, Bangkok, Thailand.
8. **Patpicha Arunsan**, Chalermlap Donthaisong, Kulwadee Suwannatrai, Sattrachai Prasopdee, Supawadee Piratae, Jutarat Kulsantiwong, Panita Khampoosa, and Smarn Tesana. Localization of cystogenous gland in *Opisthorchis viverrini* cercariae and cyst wall formation in metacercariae by chemical staining, immunohistochemistry, and immunofluorescence. The 1st symposium of Specific Health Problem in Greater Mekong Sub-region (SHeP-GMS) on 11May 2012, Auditorium 1, Faculty of Medicine, Khon Kaen University.
9. **Patpicha Arunsan**, Kulwadee Suwannatrai, Chalermlap Donthaisong, Sattrachai Prasopdee, Supawadee Piratae, Jutarat Kulsantiwong, Panita Khampoosa, and Smarn Tesana. The effects of salinity on development of miracidia in *Opisthorchis viverrini*

eggs. Current research on opisthorchiasis, cholangiocacinoma and fascioliasis on 28-29 October 2010, UdonThani Rajabhat University.

Selected Publications

1. Wirangrong Sritongklang, Nav La, Alisa Boonsuya, **Patpicha Arunsan**, Thanawat Trasaktaweesakul, Pundit Asavaritikrai, Nathkapach Kaewpitoon Rattanapitoon, & Schawanya Kaewpitoon Rattanapitoon. (2025). Molecular identification of *Entamoeba gingivalis* and its association with periodontitis in diabetic patients: A PCR-based study targeting the ITS2 region. *International Journal of Biomedicine, 15*(3), 545–551. https://doi.org/10.21103/Article15(3)_OA15
2. Natthakamol Wakhuwathapong, Pitchayapa Janyacharoen, Nita Jungpichanwanit, Sirichai Phinsiri, Alisa Boonsuya, **Patpicha Arunsan**, Weerapat Foytong, Nathkapach Kaewpitoon Rattanapitoon, Schawanya Kaewpitoon Rattanapitoon, & Chutharat Thanchonnang. (2025). Nanochitosan-coated *Zingiber cassumunar* (Plai) for anti-inflammatory applications: Characterization and therapeutic evaluation. *International Journal of Biomedicine, 15*(3), 564–571. https://doi.org/10.21103/Article15(3)_OA18
3. **Patpicha Arunsan**, Phornphitcha Pechdee, Sirichai Phinsiri, Alisa Boonsuya, Chutharat Thanchonnang, Nav La, Nattawut Keeratibharat, Nathkapach Kaewpitoon Rattanapitoon, Schawanya Kaewpitoon Rattanapitoon. (2025). Anthelmintic activity and pathophysiological effect of Allium sativum extract-based copper nanoparticles against the carcinogenic liver fluke, *Opisthorchis viverrini*. *Journal of Parasitology Research*, 2025, Article 7058749. https://doi.org/10.1155/japr/7058749
4. Chutharat Thanchonnang, Alisa Boonsuya, Phornpitcha Pechdee, **Patpicha Arunsan**, Nav La, Nattawut Keeratibharat, Nathkapach Kaewpitoon Rattanapitoon, Wiwat Nuansing, Schawanya Kaewpitoon Rattanapitoon. (2025). Capsaicin hydrogel skin patch: Development, characterization, and safety evaluation of cytotoxicity, anti-inflammatory effects, and pain-relief applications. *International Journal of Biomedicine*, 15(3), 572–582. https://doi.org/10.21103/Article15(3)_OA19
5. Phornphitcha Pechdee, **Patpicha Arunsan**, Alisa Boonsuya, Chutharat Thanchonnang, Sirichai Phinsiri, Nathkapach Kaewpitoon Rattanapitoon, Thewarach Laha, Nattawut Keeratibharat, & Schawanya Kaewpitoon Rattanapitoon. (2025). Efficacy of *Thunbergia laurifolia* crude extracts against carcinogenic liver fluke, *Opisthorchis viverrini*. *Tropical*

Biomedicine, 42(1), 65–75. https://doi.org/10.47665/tb.42.1.012

6. Phornpitcha Pechdee, Alisa Boonsuya, **Patpicha Arunsan**, Chutharat Thanchonnang, Nav La, Nathkapach Kaewpitoon Rattanapitoon, Kornmaneethana Punnasirimangmee, Pattharasedthi Pholyiam, Schawanya Kaewpitoon Rattanapitoon. Effect of *Allium sativum*, *Thunbergia laurifolia*, and *Eurycoma longifolia* crude extracts on the minute intestinal fluke, *Haplorchis taichui*. (2024). *Tropical biomedicine*. *41*(4), 543-552. https://doi.org/10.47665/tb.41.4.016
7. Phornpitcha Pechdee, Alisa Boonsuya, **Patpicha Arunsan**, Chutharat Thanchonnang, Nav La, Nathkapach Kaewpitoon Rattanapitoon, Thewarach Laha, Wiwat Nuansing, Schawanya Kaewpitoon Rattanapitoon. Anthelmintic activity and pathophysiological effect of *Allium sativum* crude extract against carcinogenic liver fluke, *Opisthorchis viverrini.* (2024). *Tropical biomedicine*, *41*(4), 427-437. https://doi.org/10.47665/tb.41.4.002
8. Alisa Boonsuya, **Patpicha Arunsan**, Phornpitcha Pechdee, Nav La, Chutharat Thanchonnang, Nathkapach Kaewpitoon Rattanapitoon, Schawanya Kaewpitoon Rattanapitoon. Diagnosis of the carcinogenic liver fluke, *Opisthorchis viverrini*: comparison of two coprological methods versus the automatic feces analyzer. (2024). *Tropical biomedicine*, *41*(3), 264–270. https://doi.org/10.47665/tb.41.3.005
9. Alisa Boonsuya, Phornpitcha Pechdee, **Patpicha Arunsan**, Chutharat Thanchonnang, Sirichai Phinsiri, Nathkapach Kaewpitoon Rattanapitoon, Schawanya Kaewpitoon Rattanapitoon. Papaya seed extract combined lauryl glucoside against human parasites contaminated in fresh vegetables. (2025). *The Natural Products Journal*, 15, e22103155322731. https://doi.org/10.2174/0122103155322731240822075823
10. Ampas Wisetmora, Oranard Wattanawong, Adulsak Wijit, Jutikarn Phukowluan, Ampol Nachairan, Prueksarawuth Jaksuay, Sivapong Sungpradit, Nuttapon Ekobol, Thidarut Boonmars, Alisa Boonsuya, Phornphitcha Pechdee, Chutharat Thanchonnang, Nav La · Nathkapach K. Rattanapitoon, **Patpicha Arunsan**, Schawanya K. Rattanapitoon. (2024). Gastrointestinal Helminthic Infection among the Population in Northern Thailand. *Acta Parasitologica*, 69(3), 1648-1660. https://doi.org/10.1007/s11686-024-00892-1
11. Warunchalee Sangkam, **Patpicha Arunsan**, Phornphitcha Pechdee, Alisa Boonsuya, Chutharat Thanchonnang, Wassana Chatdumrong, Nathkapach Kaewpitoon Rattanapitoon, & Schawanya Kaewpitoon Rattanapitoon. (2024). Anthelmintic activity and pathophysiological effect of anthelmintic drugs against carcinogenic liver fluke,

Opisthorchis viverrini. *Tropical biomedicine*, 41(2), 196-205. https://doi.org/10.47665/tb.41.2.010. (2024). Anthelmintic activity and pathophysiological effect of anthelmintic drugs against carcinogenic liver fluke*, Opisthorchis viverrini*. *Tropical biomedicine*, *41*(2), 196-205. https://doi.org/10.47665/tb.41.2.010

12. Alisa Boonsuya, Pantip Chitpitaklert, Phornpitcha Pechdee, Wirangrong Srithongklang, Chutharat Thanchonnang, Nav La, Chadaporn Gordon, Nathkapach Kaewpitoon Rattanapitoon, **Patpicha Arunsan**, Schawanya Kaewpitoon Rattanapitoon. Oral parasitic protozoan Entamoeba gingivalis in periodontal disease patients, northeastern Thailand. (2023). *Tropical biomedicine*, *40*(4): 471-477.
13. Nav LA, Monica Leng, **Patpicha Arunsan**, Phornpitcha Pechdee, Alisa Boonsuya, Chutharat Thanchonnang, Nathkapach Kaewpitoon Rattanapitoon, Schawanya Kaewpitoon Rattanapitoon. Molecular identification of Opisthorchis viverrini among the northeastern Cambodian population by internal transcribed spacer 2 based polymerase chain reaction. (2023). *Tropical biomedicine*, *40*(4): 383-391.
14. Pantip Chitpitaklert, Alisa Boonsuya, Phornpitcha Pechdee, Chutharat Thanchonnang, Nav LA, Nathkapach Kaewpitoon Rattanapitoon, **Patpicha Arunsan**, Schawanya Kaewpitoon Rattanapitoon. Molecular detection of oral Trichomonas tenax in periodontal disease patients by polymerase chain reaction -based 18S rRNA gene. (2023). *Tropical biomedicine*, *40*(3): 307-312.
15. Nav LA, Nathkapach Kaewpitoon Rattanapitoon, Phornpitcha Pechdee, Alisa Boonsuya, **Patpicha Arunsan**, Schawanya Kaewpitoon Rattanapitoon. Intestinal parasitic infections and risk factors among the population in Cambodia. (2022). *Tropical biomedicine*, *39*(4):539-546.
16. Sujittra Chaiyadet, Sirikachorn Tangkawattana, Michael J Smout, Wannaporn Ittiprasert, Victoria H. Mann, Raksawan Deenonpoe, **Patpicha Arunsan**, Alex Loukas, Paul J Brindley, Thewarach Laha. Knockout of liver fluke granulin, Ov-grn-1, impedes malignant transformation during chronic infection with *Opisthorchis viverrini.* (2022). *PLoS Pathogens*, 18(9), e1010839.
17. **Patpicha Arunsan**, Apisit Chaidee, Christina J. Cochan1, Victoria H. Mann, Toshihiko Tanno,Chutima Kumkhaek, Michael J. Smout, Shannon E. Karinshak, Rutchanee Rodpai, Javier Sotillo, Alex Loukas, Thewarach Laha, Paul J. Brindley, Wannaporn Ittiprasert. Liver fluke granulin promotes extracellular vesicle-mediated crosstalk and cellular

microenvironment conducive to cholangiocarcinoma. (2020). *Neoplasia*, 22: 203-216.

18. **Patpicha Arunsan**, Wannaporn Ittiprasert, Michael J Smout, Christina J Cochran, Victoria H Mann, Sujittra Chaiyadet, Shannon E Karinshak, Banchob Sripa, Neil David Young, Javier Sotillo, Alex Loukas, Paul J Brindley, Thewarach Laha. (2019). Programmed knockout mutation of liver fluke granulin attenuates virulence of infection-induced hepatobiliarymorbidity. *eLIFE*, 8: e41463: 1-24.
19. Kulwadee Suwannatrai, Apiporn T Thinkhamrop Suwannatrai, Chalermlap Donthaisong, **Patpicha Arunsan**, Kavin Thinkhamrop, Pairat Tabsripair, Jariya Umka Welbat, Sirikachorn Tangkawattana, Javier Sotillo, Smarn Tesana. Hemocyte subpopulation changes in Bithynia snails infected with *Opisthorchis viverrini* in Thailand. (2019). *BioRxiv*, 1-23.
20. Chalermlap Donthaisong, **Patpicha Arunsan**, Kulwadee Suwannatrai, Sattrachai Prasopdee, Jutharat Kulsantiwong, Sutee Wongmaneeprateep, Apiporn Suwannatrai and Smarn Tesana. (2016). Infectivity and development of *Opisthorchis viverrine* metacercariae in the immunosuppressed cyprinid fish fingerlings, *Barbonymus gonionotus*. *Acta Tropica*, 162:107-113.
21. Chalermlap Donthaisong, **Patpicha Arunsan**, Kulwadee Suwannatrai, Sattrachai Prasopdee, Jutharat Kulsantiwong, Sutee Wongmaneeprateep, Apiporn Suwannatrai, Smarn Tesana(2014). Experimental infection of *Opisthorchis viverrini* cercariae to the cyprinid fish,*Barbonymus gonionotus*. *Acta Tropica*, 136: 118-122.
22. **Patpicha Arunsan**, Chalermlap Donthaisong, Kulwadee Suwannatrai, Sattrachai Prasopdee, Jutharat Kulsantiwong, Apiporn Suwannatrai, Thidarut Boonmars, Jariya Umka Welbat,Songpol Oopachitakul and Smarn Tesana. (2014). Localization of the cystogenous glands of *O. viverrini* cercariae. *Southeast Asian J Trop Med Public Health*, 45: 276-289.